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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	Explainable Graph Neural Network Framework for Supply Chain Risk Prediction in Multi-Tier Supply Chain Networks using Big Data Processing	
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Abstract

Modern supply chains involve complex relationships among products, production activities, factory operations, storage facilities, and deliveries. Identifying products that may be exposed to operational risk is important for effective supply chain monitoring and decision-making. Traditional machine learning models generally treat products as independent records and may not capture relationships between products that share common supply chain characteristics.

This project proposes an Explainable Graph Neural Network (GNN) Framework for Supply Chain Risk Prediction in Multi-Tier Supply Chain Networks using Big Data Processing. The framework integrates multiple supply chain datasets containing product-level information related to sales, production, factory issues, and delivery operations. These datasets are cleaned, integrated, and transformed into product-level operational features using data preprocessing and feature engineering techniques. PySpark is incorporated to demonstrate scalable processing of supply chain data.

A product-centric graph is constructed in which each product is represented as a node, while relationships between products derived from common plant, storage, and product-group associations are represented as edges. Operational indicators such as production efficiency, delivery efficiency, factory issue ratio, sales variability, production variability, and operational gaps are used to characterize the risk behaviour of each product. A composite risk score is generated and products are categorized into **low-, medium-, and high-risk classes**.

The resulting graph is processed using PyTorch Geometric, and a Graph Convolutional Network (GCN) is trained to predict the risk category of each product by learning both its operational characteristics and the relationships with connected products. To improve interpretability, GNNExplainer is used to identify important features and graph information contributing to the model's predictions. These explanations are converted into human-readable reasons such as unstable sales demand, high factory issues, production inefficiency, and delivery-related problems.

The proposed framework is evaluated using accuracy, precision, recall, and F1-score and compared with conventional machine learning models such as Random Forest and XGBoost. The project aims to demonstrate how combining big-data preprocessing, graph-based learning, and explainable AI can provide an interpretable product-level supply chain risk prediction system that can support operational monitoring, internal auditing, inventory planning, and supply chain decision-making.

Signature of the Guide

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